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# Student Registration Form

**Aim:-** To Develop a student Registration form using Java Swing Components such as text fields, radio buttons, & checkboxes, with basic input validation.

**Theory:-**

- Java swing is a Java library used to create graphical user Interface (GUI)
- Components like JLabel, JTextfield, JButton, JRadioButton & JCheckBox helps create an interactive registration form.
- Action listener handles button-click event & JOptionPane Display messeges to user.

**Code:-**

```
import javax.swing.*;
import java.awt.event.*;
public class StudentRegistration extends JFrame
        implements ActionListener {
    JTextfield name, roll, branch;
    JRadioButton male, female;
    JCheckBox terms;
    JButton Studen submit;

StudentRegistration() {
    setTitle("Student Registration");
    setSize(350, 300);
    setLayout(null);
```

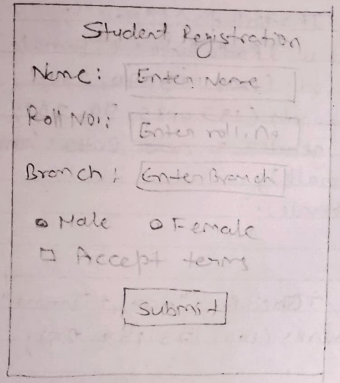
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```
addLabel("Name :", 30, 30);
name = addField(120, 30);
addLabel("Roll No. :", 30, 70);
roll = addField(120, 70);
addLabel("Branch :", 30, 110);
branch = addField(120, 110);

male = new JRadioButton("Male");
female = new JRadioButton("Female");
male.setBounds(120, 145, 70, 25);
female.setBounds(195, 145, 90, 25);
ButtonGroup gender = new ButtonGroup();
gender.add(male);
gender.add(female);
add.male;
add.female;
terms = new JCheckBox("Accept Terms");
terms.setBounds(100, 175, 150, 25);
add(terms);
Submit = new JButton("Submit");
submit.setBounds(110, 210, 100, 30);
submit.addActionListener(this);
add(submit);
setDefaultCloseOperation(EXIT_ON_CLOSE);
setVisible(true);
}

void addLabel(String text, int x, int y){
    Jlabel label = new Jlabel(text);
    label.setBounds(x, y, 90, 25);
    add(label);
}
```

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```
JTextField addField(int x, int y){
    JTextField field = new JTextField();
    field.setBounds(x, y, 160, 25);
    add(field);
    return field;
}
@Override
public void actionPerformed(ActionEvent e){
    if(name.getText().trim().isEmpty()
        || roll.getText().trim().isEmpty()
        || branch.getText().trim().isEmpty()){
    JOptionPane.showMessageDialog(this,
        "Please fill all fields");
} else if(!male.isSelected() &&
        !female.isSelected()){
    JOptionPane.showMessageDialog(this,
        "Select Gender");
} else if(!terms.isSelected()){
    JOptionPane.showMessageDialog(this,
        "Accept terms & conditions");
} else {
    JOptionPane.showMessageDialog(this,
        "Registration successful!");
    }
}

public static void main(String[] args){
    new StudentRegistration();
}
}
```

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## Viva Questions

**Q1** What is Java swing?

**Ans** Java swing is a java library used to create graphical user interfaces with components such as buttons, labels, & text fields.

**Q2** What is the use of JFrame?

**Ans** JFrame creates the main application window in a Swing program.

**Q3** What is the use of JTextField?

**Ans** JTextfield allows the user to enter a single line of text.

**Q4** Why is ButtonGroup used?

**Ans** ButtonGroup ensures that only one radio button in a group can be selected at a time.

**Q5** What is the use of ActionListener?

**Ans** ActionListener handles action events, such as clicking the submit button.

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# MVC

**Aim:-** To design & develop a calculator using Java Swing that performs basic arithmetic operations using the MVC (Model - View - architecture

**Theory**

MVC - stands for Model, View & Controller. It seperates an application into three parts.

1) Model - Performs calculations such as addition, substraction, multiplication & division.
2) View - Provides the graphical interface using swing Components such as JFrame, JTextField & JButton.
3) Controller - Handles button-click events, takes input from the View, calls the Model & displays the result.

This separation makes the program easier to understand, maintain & modify.

**Code :-**

File 1 CalculatorModel.java

```
public class CalculatorModel {
    public double Calculate(double a, double b,
                            String op) {
        switch (op) {
        case "+": return a+b;
        case "-": return a-b;
        case "*": return a*b;
        case "/":
```

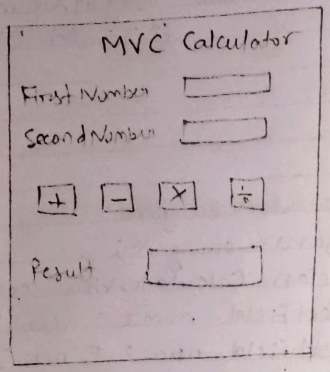
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```
        if(b==0)
            throw new ArithmeticException("Cannot
        deride by zero);
            return a/b;
        default;
            throw new IllegalArgumentException
                        ("Invalid Operator");
        }
    }
}
```

File 2 CalculatorView.java

```
import javax.swing.*;
public class CalculatorView extends JFrame{
    JTextField num1 = new JTextField();
    JTextfield num2 = new JTextField();
    JTextfield result = new JTextField();

    JButton add = new JButton("+");
    JButton sub = new JButton("-");
    JButton mul = new JButton("*");
    JButton div = new JButton("/");
public CalculatorView(){
    setTitle("MVC Arch Calculator");
    setSize(350,280);
    setLayout(null);
    addLable("First Number:",30,30);
    num1.setBounds(140,30,150,25);
    add(num1);
```




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```
addLabel("Second number :", 30, 70);
num2.setBounds(140, 70, 150, 25);
add(num2);
add.setBounds(30, 120, 60, 30);
add.setBounds(100, 120, 60, 30);
add.setBounds(170, 120, 60, 30);
add.setBounds(240, 120, 60, 30);

add(add);
add(sub);
add(mul);
add(div);
addLabel("Result : ", 30, 100);
result.setBounds(140, 180, 150, 25);
result.setEditable(false);
add(result);
setDefaultCloseOperation(EXIT_ON_CLOSE);
setVisible(true);
}

void addLabel(String text, int x, int y){
    JLabel label = new JLabel(text);
    label.setBounds(x, y, 110, 25);
    add(label);
    }

}
```

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File 3 CalculatorController.java,

```
import java.awt.event.*;
import javax.swing.*;

public class CalculatorController implements
ActionListener {
    CalculatorModel model;
    CalculatorView view;

    public CalculatorController (CalculatorModel m,
    CalculatorView v) {
        model = m;
        view = v;
    view.add.addActionListener (this);
    view.sub.addActionListener (this);
    view.mul.addActionListener (this);
    view.div.addActionListener (this);
    }
public void actionPerformed( ActionEvent e ) {
    try {
        double a = Double.parseDouble (view.num1.getText());
        double b = Double.parseDouble (view.num2.getText());
        String op = e.getActionCommand ();
        double ans = model.calculate (a, b, op);
        view.result.setText (String.valueOf (ans));
    } catch (NumberFormatException ex) {
        JOptionPane.showMessageDialog (view,
            "Please enter valid number"); }
    catch(ArithmeticException ex){
        JOptionPane.showMessageDialog (view, ex.getMessage
                                        ());
    }
}
```

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File 4. Main.java.

```
public class Main {
    public static void main(String[] args) {
        CalculatorModel model = new CalculatorModel();
        CalculatorView view = new CalculatorView();
        new CalculatorController(model, view);
    }
}
```

Viva Question

Q1 What is MVC architecture?

Ans MVC stands for Model - View - Controller. It seperates an application into data/logic, user interface & event handling components.

Q2 What is the role of the Model?

Ans The model contains the calculation logic & performs arithmetic operations such as addition, substraction, multiplication & division.

Q3 What is the role of View?

Ans The view displays the calculator interface using swing components such as JFrame, JTextfield & JButton.

Q4 What is the role of Controller?

Ans The controller handles button-click events, read input values, calls the models & Displays the calculated result.

Q5 Why do we use Java Swing?

Ans Java Swing is used to create graphical user interfaces with components such as buttons, labels, text fields & windows.

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Aim:- To develop a Java program using JDBC to connect to a MySQL database, insert student records & retrieve & display the records.

Theory:-

→ JDBC stands for Java Database Connectivity. It allows java programs to communicate with databases such as MySQL.

→ The connection interfaces establishes a connection with the database, PreparedStatement executes parameterized SQL queries & ResultSet stores the results of a SELECT Query.

→ JDBC requires a suitable database driver, such as MySQL connector/J, to communicates with MySQL.

Code:-

Step 1 Create the database & table in MySQL

```
CREATE DATABASE College;
USE college;
CREATE TABLE student (
    id INT PRIMARY KEY,
    name VARCHAR(30),
    branch VARCHAR(20));
```

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Step 2 Java program - JDBCDemo.java.

```
import java.sql.*;
public class JDBCDemo {
    public static void main(String[] args) {
    String url = "jdbc:mysql://localhost:3306/college";
    String user = "root";
    String password = "root";

try (Connection con = DriverManager.getConnection
            (url, user, password));{
    String sql = "INSERT INTO student VALUES
                    (?, ?, ?)";
try( PreparedStatement ps = con.prepareStatement
                    (sql)) {

    ps.setInt(1, 101);
    ps.setString(2, "Rahul");
    ps.setString(3, "CSE");
    ps.executeUpdate();
}
System.out.println("Record Inserted");
try( Statement st = con.createStatement();
    ResultSet rs = st.executeQuery("SELECT *
                    FROM student "));{
while (rs.next()) {
    System.out.println(
     rs.getInt("id") + " " +
     rs.getString("name") + " " +
     rs.getString("branch"));
    }
}
```

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Output

Record Inserted.
101 Rahul CSE

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```
        }
        catch (SQLException e) {
            System.out.println(e.getMessage());
        }
    }
}
```

Viva Question

Q1 What is JDBC ?
Ans JDBC stands for Java database connectivity. It allows Java programs to connect to & interact with database

Q2 What is a JDBC driver ?
Ans A JDBC driver enables Java to communicate with a database. MySQL Connectivity/J is the driver used for MYSQL.

Q3 What is the use of Connection ?
Ans Connection represents an active connection between a Java application & a database.

Q4 What is the difference between Statement & preparedStatement ?
Ans Statement executes ordinary SQL queries, while PreparedStatement uses parameterized queries with placeholder (?).

Q5 What is ResultSet ?
Ans ResultSet stores the rows returned by a SQL query & allows the program to access them one by one using next().

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